ADAPTIVE SOFTWARE ENGINEERING · 24CI3201 · HACKATHON

ReqGenie

An adaptive AI system that turns client requirements into user stories, acceptance criteria and UML diagrams

Team 17 · Section S11

THE PROBLEM

Requirements are slow and messy

Unclear input

Client needs are scattered across emails, calls and meeting notes.

Slow manual work

Writing user stories and UML diagrams by hand takes days.

Constant changes

Every change means redoing stories and diagrams again.

OUR SOLUTION

ReqGenie turns client words into a ready plan

User stories

Clear “As a user, I want…” stories.

Acceptance criteria

Simple checks to confirm each story is done.

UML diagrams

Use case, class and sequence diagrams.

HOW IT WORKS

Five simple steps

1. **Input**
   Paste client notes
2. **Analyse**
   AI finds actors and needs
3. **Stories**
   Stories and criteria
4. **Diagrams**
   UML drawn automatically
5. **Export**
   PDF or JIRA cards

One input, one click, complete requirement documents.

WHAT IT DOES

Key features

Easy input

Paste text or upload meeting notes.

Story generator

User stories with acceptance criteria.

Priority tags

Must, Should, Could, Won't (MoSCoW).

UML diagrams

Use case, class and sequence diagrams.

Adaptive updates

Re-generates only what a change affects.

Easy export

Download as PDF or JIRA-style cards.

THE ADAPTIVE PART

When requirements change, ReqGenie adapts

1. Change

Client edits or adds a requirement.

2. Detect

AI finds which stories are affected.

3. Update

Only those stories and diagrams are redone.

Faster response to change, with a saved version history.

SYSTEM DESIGN

Simple three-part architecture

Front end

- Input form
- Stories viewer
- Diagram viewer

Back end + AI

- Language analysis
- Story generator
- Change detector

Output

- UML renderer
- Version database
- PDF export

TECH STACK

Tools we will use

Layer	Technology
Front end	React, HTML, CSS
Back end	Python with FastAPI
AI and language	LLM API, spaCy
Diagrams	PlantUML or Mermaid
Database	MongoDB or SQLite
DevOps	Git, Docker, JIRA board

OUR TEAM

Team 17 and roles

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Front end and UI

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AI and back end

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Diagrams, export and testing

AGILE PLAN

Four sprints to the demo

Sprint	Goal
Sprint 1	Requirements, UI design, project setup
Sprint 2	Input form, story and criteria generator
Sprint 3	UML diagrams and change detection
Sprint 4	Export, testing, final demo

We follow Scrum, track tasks on a JIRA board and use Git feature branches.

EXAMPLE

From client words to a user story

Client says

“We need an online shop where customers can search products and pay by UPI.”

ReqGenie creates

Story: As a customer, I want to search products so that I can find items quickly.

Check: Results appear in under 3 seconds.

Priority: Must have

Diagram: Use case with Customer, Search Product, Pay by UPI.

BENEFITS

Thank you

Faster planning

Fewer mistakes

Easy to change

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